

# Omar Basem Abdulsalam Muhammed El-Nahass

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## Objective

Innovative Mechatronics Technology student (B. Tech, Suez Canal University, expected 2026) specializing in mechanical design, Mechanical engineering, automation, and robotics. Proven track record in CAD modeling and Certified SOLIDWORKS Professional Advanced (CSWPA) and SOLIDWORKS Champion., manufacturing process optimization, and cross-functional project leadership. Adept at driving efficiency improvements and designing solutions to meet standards for marine and automation systems. Seeking to contribute technical expertise and leadership to global engineering teams.

## Education

**B. Tech Mechatronics Technology — Suez Canal University**, Egyptian Chinese College for Applied Technology

Graduated 2026 GPA: 3.04/4.0 (76%)

Relevant Coursework: CAD, CNC, Hydraulic & Pneumatic, Embedded Systems, PLC Programming, Manufacturing Process Design.

## Technical Experiences

**Technical Office Intern, Port Said Stadium — Port Said, Egypt | August 2025**

Conducted QC inspections for electromechanical installations, achieving 100% project sign-off

Supported site management for elevator integration and electrical safety audits.

**Mechatronics Engineer Intern – Suez Canal Authority Port said Shipyard, Port said, Egypt | July 2025**

Trained at production, Ship building & Electrical Sectors at shipyard, have developed much experiences in mechanical production processes machining heat treatment piping and production management.

**Summer Internship at Beijing Information Technology College, Beijing, China (BITC) | June 2024**

Mastered NAO robotics programming and drone piloting Gained proficiency in 3D printing workflows and MATLAB-based system simulations.

## Projects

**Graduation Project - Unmanned Surface Vessel:** Managed end-to-end mechanical design and manufacturing of a catamaran hull, integrating cyclonic filtration and navigation systems. Led a multidisciplinary team through full-scale system integration and operational commissioning for offshore oil spill recovery.

**Steering Systems:** Designed and manufactured two steering systems for Formula Student and EVER Egypt racing vehicles; increased responsiveness measurably.

**Cold Rolling Machine Prototype:** Managed end-to-end design, manufacturing, and commissioning of metal rolling systems for academic research.

**ROV (Remotely Operated Vehicle):** Led hull sealing and testing for competition-grade underwater ROV; achieved full operational status ahead of schedule.

## Courses

**Mechatronics, Intelligence and Digitalization in Green Hydrogen Systems (Coventry University UK, British University in Egypt) |** March 2026 – Jan 2027.

**Manufacturing Process Selection (Udemy) |** Nov 2024

**SolidWorks Fluid Dynamics Analysis (Udemy) |** Feb 2024

**Autodesk CAD/CAM/CAE for Mechanical Engineers (Coursera) |** Oct–Dec 2023.

## Technical Skills

**Core Competencies:** Mechanical Design, CAD/CAM, FEA, CFD Rapid Prototyping, DFM, Manufacturing Drawings, CNC Programming & Operating (Siemens/GSK)

**Software & Tools:** SOLIDWORKS, Fusion 360, AutoCAD, Rhinoceros 7, ANSYS, MAXSURF, Proteus, TIA Portal, EKTS, Cura Slicer, Cypcut, Laserworks, GstarCAD, Draftsight.

**Programming:** C, MATLAB, PLC LADDER-STL, AVR Interfacing.

## **Competitions & Awards**

**University's 8th Scientific Conference:** 1<sup>st</sup> Place - Innovation | 2026

**Collage 8th Scientific Conference:** 3<sup>th</sup> Place – Posters | 2026

**Collage 7th Scientific Conference:** 5<sup>th</sup> Place | 2025

**Rally Egypt:** 4th overall Egypt | 2024

**EVER Egypt:** 3rd place Dynamic Track & 5th Overall | 2024

**AAKRUTI Global:** 2024, 2025.

## **Volunteering Experience**

**Technical Team Leader, Robotics Head, Eccatronics Team** | Oct 2025

**Vice President, Rally Society** | Aug 2025

**Head of Steering Committee, SCU Racing Team** | Jan–Oct 2025

**Head of Mechanical Committee, Eccatronics Team** | Oct 2024 – Oct 2025

**IEEE-SCU Mechanical Design Member** | Oct 2023-Oct 2025

Other roles: Steering Committee Member, ROV Mechanical Committee Member (2023–2025); demonstrated advanced team coordination and mentoring.

## **Languages**

**Arabic:** Native.

**English:** B2.

**Chinese:** A1 – HSK 1 Certified.